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Conference 2000



Overview: New Palm OS® Technologies

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Palm OS® 4.0

- Next release of Palm OS®: version 4.0
- We're telling you all about it, in advance
- You can get a beta copy today
- You can start writing code for the new features
  - (Or fix bugs that the new debug rom finds for you)

Biggest New Features in 4.0

- Secondary storage and expansion
- Telephony
- Web Clipping everywhere
- Bluetooth ready
- Many general improvements
  - Attention Mgr
  - Exchange Libraries
  - Locale modules
  - 24-bit color APIs, Time Zones, Security…

Platform Versus Product

- Potentially flashable update for existing devices, and for new ones
  - Each device manufacturer controls timing and method of the release to their customers
- Not every feature will be on every device
- Platform provides base upon which licensees, Palm, and developers can all innovate
- Still in development; products expected during the first half of next year

What It Isn't

- PIM Data or other compatibility break
- Desktop software change (sort of)
- Switch to ARM
- Full support for different screens

How to Get It

- Sign a paper license, recieve a CD
- (Tonight in the labs, and tomorrow during the day and at the labs…)

- And/or, join the seeding program in the Pavilion at www.palmos.com
  - This will get you updated versions, ongoing

Expansion in 4.0

- Most expansion devices are secondary storage, and provide a traditional filesystem
- Secondary storage: not like built-in databases
  - Different physical characteristics, so different API
  - Pretty similar user experience
- Palm OS® apps are copied into main RAM when running (just like on a desktop), and are automatically removed when they quit

Expansion—Under the Covers

- Launch codes and notifications tell you if you're running from a card, if a card is inserted and you should quit, etc.
- PRCs are in a specific directory on the card: /Palm/Launcher/
- /Palm/Start.prc is automatically run upon insertion
  - If device is locked, this is delayed
- Use POSE and HostFS for testing

Scenarios

- How it works in the simple case, if you don't have any explicit support for expansion
  - App gets copied in, launched, deleted when quit
- A few things to watch out for
  - Registering for notifications
  - Libraries or supporting PRCs
- How to test, without any hardware
- New capabilities of your application
  - Let users read and/or save data on a card
  - Take action on insertion/removal

Attention Manager

- In a wireless world, lots of things can compete for the user's attention
  - Email, pages, SMS, Bluetooth...
- Particularly for unexpected or unsolicited events, need to be careful of the user scenario
- New concept: subtle and insistent notifications
- New hardware to get the user's attention more effectively
- New software features to make life easier, and help them manage their time

Subtle Versus Insistent

- Previously, applications were always insistent and would interrupt the user; sometimes that is still appropriate
  - e.g., Meeting alarms, Low battery, Phone call…
- Sometimes it'll depend on how the user wants to use the software and/or hardware
  - e.g., Pagers
- Hardware manufacturer specifies when and how to use attention-getting features
  - Vibration, LEDs, sound…
  - Applications can override or extend

New Attention UI

- Gather multiple items into a list
- Global Snooze now possible
- “Go To” button
- “Clear All” button
- Most of the display is controlled by the application requesting attention

Which Manager Is Which

- These managers are all different from each other, and are mostly unrelated
- Attention Manager
  - Manages common UI
- Alarm Manager
  - Sends an event to an application at a specified time
- Notification Manager
  - Handles the delivery of notifications to whoever cares about that particular event

Exchange Manager and “Send”

- Until 4.0, Exchange Manager only was for IR
- Now is a general transport mechanism
  - Pagers, SMS, Bluetooth, removable cards, notification services, HTTP, FTP…
- Exchange Libraries (plug-ins) handle transport and delivery
- For applications, just a small UI change
  - Add “Send” under “Beam”
  - Add a line or two of code
- Library selection, and addressing, are handled automatically

Exchange Libraries

- Libraries register to say that they should be listed in the “Send” list
- Might register for other services, by URL scheme
  - http, mailto…
- Present UI for addressing, progress, errors
- Do the actual sending and receiving

Application Duties

- Send data, very similar to “Beam” code
- Register in order to receive data sent to them
  - By creator ID
  - By MIME type
  - By filename extension
  - By URL scheme
- Should provide a preview of received data
  - Particularly important for unsolicited and unexpected data
- Accept (put away) the received data

Exchange Manager Arbitration

- Multiple applications might have registered for the same scheme, MIME type, creator ID, etc.
- System keeps default setting, but doesn't have UI for user to adjust for received objects
- Applications should provide a way to select theirs as the default

Local Exchange Library ("Transfer")

- Visible in debug builds, for debugging use
- Always present, so can deliver data into the built-in applications easily

Security

- UI changes to encourage use, and make applications more strict about security
  - Undo masking for just one record
  - Event alarms require password to show details
  - Password hint field is provided for user
  - Can set Auto-lock at time, or after idle
- System changes to prevent easy circumvention when device is locked:
  - Can’t connect debugger (reboot or shortcut…1 & 2)
  - Can’t perform HotSync™ operation
  - Can’t receive beam
  - Delays running auto-start applications

Telephony

- Full APIs to control telephones
- Enables one-piece or two-piece solutions
- Modular approach, with phone manufacturer providing a PRC to add support for their phone
- SMS application, APIs, and Exchange Library

Bluetooth

- Not strictly speaking part of 4.0
- Will be hardware and software add-ons
- OS is Bluetooth “ready”, but it wasn’t in time for the OS release

Web Clipping

- Support for other hardware
  - Color
  - Different modems and networks
- Support for other countries (%Location)
- Support for other character sets
- More support for “unfriendly” content
- Cookies, Exchange Libraries, etc., etc…

Backward Compatibility

- Your applications should be fine in 4.0, if they worked fine in 3.5 and debug ROMs
- 16 bit leads to less available memory, probably…
- Your Stack size has been adjusted to compensate for OS's increase
- Pose and Debug ROMs flag more incorrect things

Future Compatibility

- Pose and Debug ROMs flag more incorrect things
- #defines to prevent access to structs—very highly recommended
- New Accessor functions, for the system's private structures (so we can change them)
  - BmpGetBitDepth, TblGetNumberOfColumns...

SDK and Tools

- Most existing toolchains (and applications) will work without modification
- Updates for new features
  - Include files and libraries
  - 16-bit color in Constructor, PalmRez, Simulator, WCA Builder, Palm OS® Emulator
  - Constructor can edit PRCs directly
  - New Conduit APIs for expansion cards
  - New Palm OS Emulator, for VZ, expansion, etc.
  - PRCs to help test, like HostFS for Pose

Other 4.0 Talks

- 201 is about Telephony
- 202 is Expansion (with 211 as its continuation)
- 203 is more detail on general handheld APIs
- 204 is Bluetooth
- 205 is Web Clipping
- 209 and 210 are roundtables to discuss this all (and to chat with the engineers!)

The Seeding CD

- Four Debug ROMs are on the seeding CD
  - "Kitchen sink": everything that could theoretically be run on a Palm™ V handheld
    - English, French, German, Spanish, Italian
    - Web Clipping, Expansion, Telephony
    - …Etc.
  - "VZ and 16-bit color" (debug and nondebug)
    - Plus Expansion, but not much else
  - Japanese with 8-bit color
- SDK, Pose, Documentation, HostFS.prc


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